

EVAL REPORT — Ayush Rathod AI Persona

2026-06-06 10:55 UTC · 30 golden questions · ~9 voice calls (32 turns) · answer=gpt-4.1-mini, judge=gpt-4.1-mini

0.48s FIRST AUDIO (TTS TTFB) 100% < 2s · p95 1.03s	0% HALLUCINATION RATE 0/30 questions	100% OUT-OF-CORPUS REFUSAL 4/4 · 0 fabrications	1.00 RETRIEVAL RECALL@6 precision 0.62
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VOICE QUALITY

First audio out — TTS TTFB (p50 / p95)	0.48s / 1.03s [<2s: 100%]
Full-turn E2E — stops to speaks (p50 / p95)	4.57s / 8.14s [<2s: 0%]
Per-stage p50 — RAG retrieval / LLM TTFT / TTS	1.30s / 0.88s / 0.48s
Booking success rate	7/10 (70%) [manual]
Transcription proxy WER	n/a (transcript log unavailable)
E2E is the full perceived turn; dominated by cross-region RAG (~1.3s, in the critical path) + semantic end-of-turn wait, not synthesis. First audio (0.48s) is the <2s figure.	

CHAT GROUNDEDNESS

Hallucination rate	0% (0/30 questions)
Retrieval precision@6 / recall@6	0.62 / 1.00
Out-of-corpus refusal rate (end-to-end)	4/4 (100%)
Answer correctness / mean relevance	88.5% / 0.61
Retrieval gating let 4/4 out-of-corpus queries clear the floor (resume 'name-magnet'); the persona declined all of them on content — honesty holds at the generation layer.	

FAILURE MODES

- Cross-region RAG in the critical path** — full-turn E2E measured 4.57s p50 while first audio (TTS TTFB) is 0.48s; retrieval alone adds ~1.3s (embed + Qdrant, cross-region). Fix: prewarm the embed+Qdrant TLS connection during the greeting (cold first turn ~2s -> ~0.9s); speculative retrieval during caller speech is the next step.
- Out-of-corpus 'name-magnet'** — the resume contact chunk scores 0.5-0.66 for any query mentioning Ayush, so retrieval low-confidence gating fired on 0/4 out-of-corpus questions. Fix: per-chunk relevance floor + a persona that declines on content kept end-to-end refusal at 100%; a cross-encoder reranker is the deeper fix.
- Multi-source synthesis miss** — precision drops to 0.47 (project) / 0.63 (fit) vs 0.73-1.0 on factual lookups, and all 3 correctness misses were grounded-but-incomplete multi-part answers (named one source, omitted a second) with 0 hallucinations. Fix: a specificity nudge was tested and reverted (it induced a fabricated stat) - groundedness over completeness; per-accomplishment resume chunking is the deeper fix.

TRADEOFF

Cascade (STT -> LLM -> TTS) over speech-to-speech. It costs an extra hop of latency, but it is the only way to inject retrieved RAG context as text, to measure hallucination and retrieval quality per stage for this report, and to call the Cal.com booking tools reliably. Streaming TTS holds first audio at 0.48s p50 (under the 2s budget); the accepted cost is full-turn E2E sitting above 2s under grounded retrieval.

WITH 2 MORE WEEKS

- Speculative retrieval during caller speech** — fire the embed+Qdrant round trip on the partial transcript before end-of-utterance, hiding the ~1.3s retrieval RTT behind the caller's trailing silence (the highest-leverage voice-latency item).
- Cross-encoder reranker over the ~20 candidates already over-fetched** — reorders the out-of-corpus 'name-magnet' and low-signal chunks below real resume content; ship it on the chat path first where the latency budget is relaxed.
- Hybrid dense+BM25 + proper WER** — sparse matching for exact model names / version numbers / metric figures (the under-specificity class), plus recorded test calls with user_transcript persisted onto TurnMetrics so transcription accuracy stops being n/a.

GLOSSARY

Abbreviations used in this report (page 1 is the complete measurement deliverable).

ABBREVIATIONS	
STT	Speech-to-Text — transcribes the caller's audio into text (Sarvam Saaras:v3).
LLM	Large Language Model — generates the persona's reply (GPT-4.1-mini).
TTS	Text-to-Speech — synthesises the spoken reply, streamed back to the caller (Sarvam Bulbul:v3).
RAG	Retrieval-Augmented Generation — answers are grounded in retrieved resume / GitHub text, not model memory.
TTFB	Time to First Byte — here, the delay until the first audio chunk reaches the caller. This is the brief's <2s first-response figure.
TTFT	Time to First Token — the delay until the LLM emits its first output token.
E2E	End-to-End — the full perceived turn: caller stops speaking until the agent starts speaking.
p50 / p95	50th / 95th percentile — the median and the tail (near-worst-case) value across all measured turns.
precision@k / recall@k	Retrieval quality over the top-k chunks: precision = fraction of retrieved chunks that are relevant; recall = fraction of all relevant chunks that were retrieved.
WER	Word Error Rate — a transcription-accuracy measure (reported as a proxy here; true WER needs paired ground-truth transcripts).
RTT	Round-Trip Time — one network request-and-response cycle, e.g. an embedding call or a Qdrant query.
BM25	Best Matching 25 — a classic sparse keyword-ranking function; would pair with dense vectors in hybrid search.